

Q1: 24 Feb (Shift 1) - Single Correct

The function $f(x) = \frac{4x^3 - 3x^2}{6} - 2 \sin x + (2x - 1) \cos x$:

(1) increases in $\left[\frac{1}{2}, \infty\right)$

(2) decreases $\left(-\infty, \frac{1}{2}\right]$

(3) increases in $\left(-\infty, \frac{1}{2}\right]$

(4) decreases $\left[\frac{1}{2}, \infty\right)$

Q2: 24 Feb (Shift 1) - Single Correct

If the tangent to the curve $y = x^3$ at the point $P(t, t^3)$ meets the curve again at Q , then the ordinate of the point which divides PQ internally in the ratio 1: 2 is :

(1) $-2t^3$

(2) $-t^3$

(3) 0

(4) $2t^3$

Q3: 24 Feb (Shift 1) - Numerical

The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1 - \sin x} = \alpha$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is

Q4: 24 Feb (Shift 2) - Single Correct

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4 \end{cases}$$

Let $A = \{x \in \mathbf{R} : f \text{ is increasing}\}$. Then A is equal to :

(1) $(-5, -4) \cup (4, \infty)$

(2) $(-5, \infty)$

(3) $(-\infty, -5) \cup (4, \infty)$

(4) $(-\infty, -5) \cup (-4, \infty)$

Q5: 24 Feb (Shift 2) - Single Correct

If the curve $y = ax^2 + bx + c$, $x \in R$, passes through the point (1,2) and the tangent line to this curve at origin is $y = x$, then the possible values of a, b, c are :

(1) $a = 1, b = 1, c = 0$

(2) $a = -1, b = 1, c = 1$

(3) $a = 1, b = 0, c = 1$

(4) $a = \frac{1}{2}, b = \frac{1}{2}, c = 1$

Q6: 24 Feb (Shift 2) - Single Correct

For which of the following curves, the line $x + \sqrt{3}y = 2\sqrt{3}$ is the tangent at the point $\left(\frac{3\sqrt{3}}{2}, \frac{1}{2}\right)$?

(1) $x^2 + 9y^2 = 9$

(2) $2x^2 - 18y^2 = 9$

(3) $y^2 = \frac{1}{6\sqrt{3}}x$

(4) $x^2 + y^2 = 7$

Q7: 25 Feb (Shift 1) - Single Correct

If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered pair (a, b) is equal to :

(1) $(-5, 8)$

(2) $(5, 8)$

(3) $(5, -8)$

(4) $(-5, -8)$

Q8: 25 Feb (Shift 1) - Single Correct

If the curves, $\frac{x^2}{a} + \frac{y^2}{b} = 1$ and $\frac{x^2}{c} + \frac{y^2}{d} = 1$ intersect each other at an angle of 90° , then which of the following relations is true?

(1) $a + b = c + d$

Questions with Answer Keys

MathonGo

(2) $a - b = c - d$

(3) $ab = \frac{c+d}{a+b}$

(4) $a - c = b + d$

Q9: 25 Feb (Shift 1) - Numerical

Let $f(x)$ be a polynomial of degree 6 in x , in which the coefficient of x^6 is unity and it has extrema at $x = -1$ and $x = 1$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = 1$, then 5. $f(2)$ is equal to

Q10: 25 Feb (Shift 2) - Single Correct

The shortest distance between the line $x - y = 1$ and the curve $x^2 = 2y$ is:

(1) $\frac{1}{2}$

(2) 0

(3) $\frac{1}{2\sqrt{2}}$

(4) $\frac{1}{\sqrt{2}}$

Q11: 25 Feb (Shift 2) - Numerical

If the curves $x = y^4$ and $xy = k$ cut at right angles, then $(4k)^6$ is equal to

Q12: 26 Feb (Shift 1) - Single Correct

The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point:

(1) (2,9)

(2) (2,2)

(3) $\left(3, \frac{21}{2}\right)$

(4) (0,0)

Q13: 26 Feb (Shift 2) - Single Correct

The triangle of maximum area that can be inscribed in a given circle of radius 'r' is:

Questions with Answer Keys

MathonGo

(1) A right angle triangle having two of its sides of length $2r$ and r .

(2) An equilateral triangle of height $\frac{2r}{3}$.

(3) An isosceles triangle with base equal to $2r$.

(4) An equilateral triangle having each of its side of length $\sqrt{3} r$.

Q14: 26 Feb (Shift 2) - Numerical

Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2} b = 3$, then $(a^2 + b^2 + ab)$ is equal to

Q15: 26 Feb (Shift 2) - Numerical

Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval $(a, a + 1)$. Then, $|a|$ is equal to

Answer Key**Q1** (1)**Q2** (1)**Q3** (9)**Q4** (1)**Q5** (1)**Q6** (1)**Q7** (2)**Q8** (2)**Q9** (144)**Q10** (3)**Q11** (4)**Q12** (2)**Q13** (4)**Q14** (9)**Q15** (2)

Q1 (1)

$$f'(x) = (2x - 1)(x - \sin x)$$

$$\Rightarrow f'(x) \geq 0 \text{ in } x \in \left[\frac{1}{2}, \infty\right)$$

Q2 (1)

Equation of tangent at P (t, t³)

$$(y - t^3) = 3t^2(x - t) \dots (i)$$

Now solve the above equation with

$$y = x^3 \dots (ii)$$

By (i) and (ii),

$$x^3 - t^3 = 3t^2(x - t)$$

$$x^2 + xt + t^2 = 3t^2$$

$$x^2 + xt - 2t^2 = 0$$

$$(x - t)(x + 2t) = 0$$

$$\Rightarrow x = -2t \Rightarrow Q(-2t, -8t^3)$$

$$\text{Ordinate of required point} = \frac{2t^3 + (-8t^3)}{3} = -2t^3$$

Q3 (9)

$$f(x) = \frac{4}{\sin x} + \frac{1}{1 - \sin x}$$

$$\text{Let } \sin x = t \quad \therefore x \in \left(0, \frac{\pi}{2}\right) \Rightarrow 0 < t < 1$$

$$f(t) = \frac{4}{t} + \frac{1}{1-t}$$

$$f'(t) = \frac{-4}{t^2} + \frac{1}{(1-t)^2}$$

$$= \frac{t^2 - 4(1-t)^2}{t^2(1-t)^2}$$

$$= \frac{(t-2(1-t))(t+2(1-t))}{t^2(1-t)^2}$$

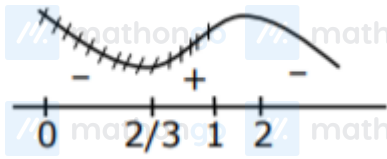
$$= \frac{(3t-2)(2-t)}{t^2(1-t)^2}$$

$$f_{\min} \text{ at } t = \frac{2}{3}$$

$$\alpha_{\min} = f\left(\frac{2}{3}\right) = \frac{4}{\frac{2}{3}} + \frac{1}{1-\frac{2}{3}}$$

$$= 6 + 3$$

$$= 9$$



Q4 (1)

$$f(x) = \begin{cases} -55 & ; x < -5 \\ 6(x^2 - x - 20) & ; -5 < x < 4 \\ 6(x^2 - x - 6) & ; x > 4 \end{cases}$$

$$f(x) = \begin{cases} -55 & ; x < -5 \\ 6(x-5)(x+4) & ; -5 < x < 4 \\ 6(x-3)(x+2) & ; x > 4 \end{cases}$$

Hence, $f(x)$ is monotonically increasing in interval $(-5, -4) \cup (4, \infty)$

Q5 (1)

$$2 = a + b + c \dots \dots (i)$$

$$\frac{dy}{dx} = 2ax + b \Rightarrow \left. \frac{dy}{dx} \right|_{(0,0)} = 1$$

$$\Rightarrow b = 1 \Rightarrow a + c = 1$$

$(0,0)$ lie on curve

$$\therefore c = 0, a = 1$$

Q6 (1)

$$\text{Tangent to } x^2 + 9y^2 = 9 \text{ at point } \left(\frac{3\sqrt{3}}{2}, \frac{1}{2} \right) \text{ is } x \times \left| \frac{3\sqrt{3}}{2} \right| + 9y \left(\frac{1}{2} \right) = 9$$

$$3\sqrt{3}x + 9y = 18 \Rightarrow x + \sqrt{3}y = 2\sqrt{3}$$

$\Rightarrow$ option (1) is true

Q7 (2)

$$f(1) = f(2)$$

$$\Rightarrow 1 - a + b - 4 = 8 - 4a + 2b - 4$$

$$3a - b = 7$$

Hints and Solutions

MathonGo

$$f'(x) = 3x^2 - 2ax + b$$

$$\Rightarrow f'\left(\frac{4}{3}\right) = 0 \Rightarrow 3 \times \frac{16}{9} - \frac{8}{3}a + b = 0$$

$$\Rightarrow -8a + 3b = -16$$

$$a = 5, b = 8$$

Q8 (2)

$$\frac{x^2}{a} + \frac{y^2}{b} = 1 \dots (1)$$

$$\text{diff: } \frac{2x}{a} + \frac{2y}{b} \frac{dy}{dx} = 0 \Rightarrow \frac{y}{b} \frac{dy}{dx} = \frac{-x}{a}$$

$$\frac{dy}{dx} = \frac{-bx}{ay} \dots (2)$$

$$\frac{x^2}{c} + \frac{y^2}{d} = 1 \dots (3)$$

$$\text{Diff: } \frac{dy}{dx} = \frac{-dx}{cy} \dots (4)$$

$$m_1 m_2 = -1 \Rightarrow \frac{-bx}{ay} \times \frac{-dx}{cy} = -1$$

$$\Rightarrow bdx^2 = -acy^2 \dots (5)$$

$$(1) - (3) \Rightarrow \left(\frac{1}{a} - \frac{1}{c}\right)x^2 + \left(\frac{1}{b} - \frac{1}{d}\right)y^2 = 0$$

$$\Rightarrow \frac{c-a}{ac}x^2 + \frac{d-b}{bd} \times \left(\frac{-bd}{ac}\right)x^2 = 0 \text{ (using 5)}$$

$$\Rightarrow (c-a) - (d-b) = 0$$

$$\Rightarrow c - a = d - b$$

Q9 (144)

$$f(x) = x^6 + ax^5 + bx^4 + x^3$$

$$\therefore f'(x) = 6x^5 + 5ax^4 + 4bx^3 + 3x^2$$

Roots 1 & -1

$$\therefore 6 + 5a + 4b + 3 = 0 \text{ \& } -6 + 5a - 4b + 3 = 0 \text{ solving}$$

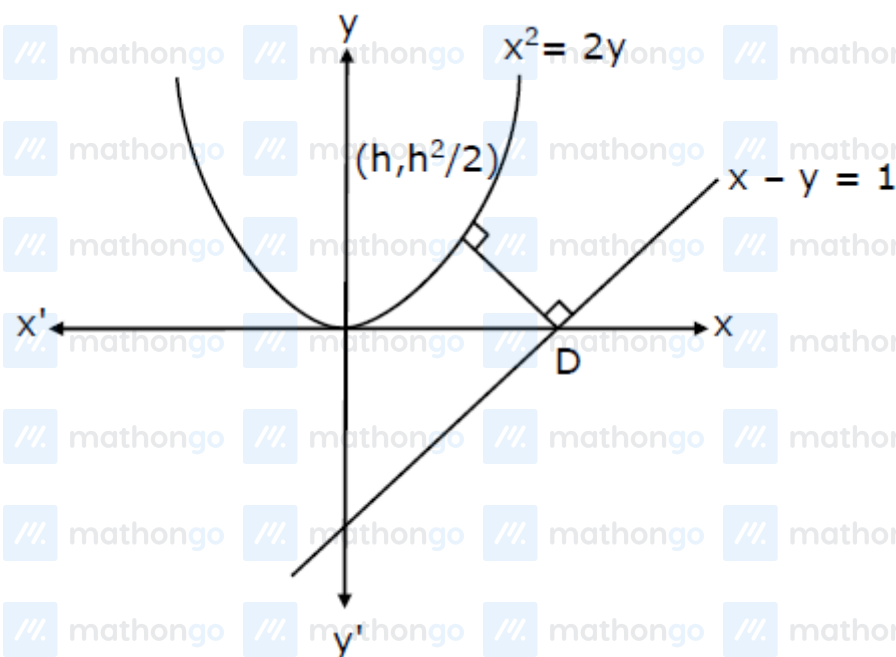
$$a = -\frac{3}{5} \quad b = -\frac{3}{2}$$

$$\therefore f(x) = x^6 - \frac{3}{5}x^5 - \frac{3}{2}x^4 + x^3$$

$$\therefore 5.f(2) = 5 \left[64 - \frac{96}{5} - 24 + 8 \right] = 144$$

Q10 (3)

Shortest distance must be along common normal



m_1 (slope of line $x - y = 1$) = 1 $\Rightarrow$ slope of perpendicular line = -1 $m_2 = \frac{2x}{2} = x \Rightarrow m_2 = h \Rightarrow$ slope of normal $-\frac{1}{h}$

$$-\frac{1}{h} = -1 \Rightarrow h = 1$$

so point is $\left(1, \frac{1}{2}\right)$

$$D = \left| \frac{1 - \frac{1}{2} - 1}{\sqrt{1+1}} \right| = \frac{1}{2\sqrt{2}}$$

Q11 (4)

Hints and Solutions

MathonGo

$$4y^3 \frac{dy}{dx} = 1 \quad \& \quad x \frac{dy}{dx} + y = 0$$

$$m_1 = \frac{1}{4y^3} \frac{dy}{dx} = \frac{-y}{x} = m_2$$

$$m_1 m_2 = -1$$

$$\frac{1}{4y^3} \times \frac{-y}{x} = -1 \therefore x = y^4$$

$$\frac{1}{4y^6} = 1 \text{ and } xy = k$$

$$y^6 = \frac{1}{4}$$

$$\Rightarrow k^6 = y^{30}$$

$$\Rightarrow k^6 = \left(\frac{1}{4}\right)^5$$

$$\therefore (4k)^6 = 4^6 \times k^6 = 4$$

Q12 (2)

$$\frac{dy}{dx} = 2x^3 - 15x^2 + 36x - 19$$

$$\text{Let } f(x) = 2x^3 - 15x^2 + 36x - 19$$

$$f'(x) = 6x^2 - 30x + 36 = 0$$

$$x^2 - 5x + 6 = 0$$

$$x = 2, 3$$

$$f''(x) = 12x - 30$$

$$f''(x) < 0 \text{ for } x = 2$$

$$\text{At } x = 2$$

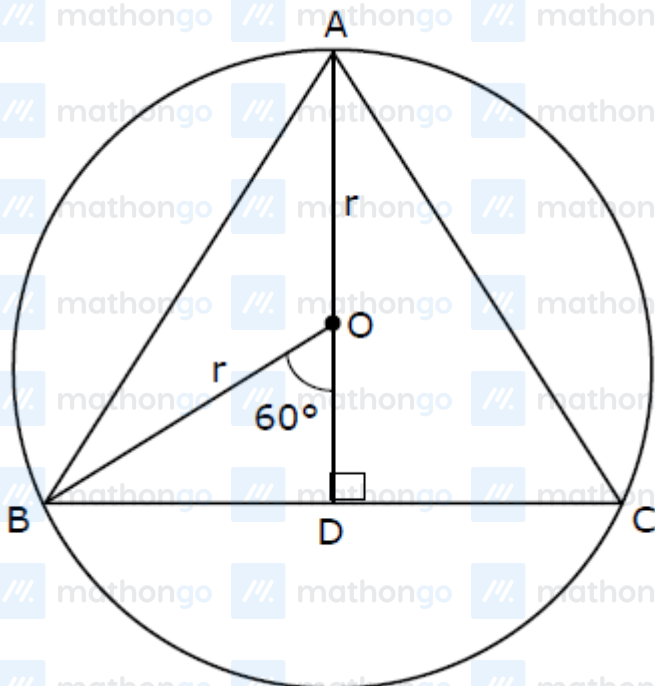
$$y = 8 - 40 + 72 - 19$$

$$y = 72 - 70 = 2$$

$$\Rightarrow (2, 2)$$

Q13 (4)

Triangle of maximum area that can be inscribed in a circle is an equilateral triangle. Let $\triangle ABC$ be inscribed in circle,



Now, in $\triangle OBD$ $OD = r \cos 60^\circ = \frac{r}{2}$

Height $= AD = \frac{3r}{2}$

Again in $\triangle ABD$ Now $\sin 60^\circ = \frac{3\frac{r}{2}}{AB}$

$\Rightarrow AB = \sqrt{3}r$

Q14 (9)

Let the equation of normal is $Y - y = -\frac{1}{m}(X - x)$, where, $m = \frac{dy}{dx}$ As it passes through (a, b)

$$b - y = -\frac{1}{m}(a - x) = -\frac{dx}{dy}(a - x)$$

$$\Rightarrow (b - y)dy = (x - a)dx$$

$$\text{by } -\frac{y^2}{2} = \frac{x^2}{2} - ax + c$$

It passes through $(3, -3)$ & $(4, -2\sqrt{2})$

$$\therefore -3b - \frac{9}{2} = \frac{9}{2} - 3a + c$$

$$\Rightarrow -6b - 9 = 9 - 6a + 2c$$

$$\Rightarrow 6a - 6b - 2c = 18$$

$$\Rightarrow 3a - 3b - c = 9$$

$$\text{Also } -2\sqrt{2}b - 4 = 8 - 4a + c$$

Hints and Solutions

MathonGo

$$4a - 2\sqrt{2}b - c = 12$$

$$\text{Also } a - 2\sqrt{2}b = 3 \quad \dots(\text{iv}) \quad (\text{given})$$

$$(\text{ii}) - (\text{iii}) \Rightarrow -a + (2\sqrt{2} - 3)b = -3 \quad \dots(\text{v})$$

$$(\text{iV}) + (\text{v}) \Rightarrow b = 0, \quad a = 3$$

$$\therefore a^2 + b^2 + ab = 9$$

Q15 (2)

$$\text{Let, } f(x) = 2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$$

$$\Rightarrow f'(x) = 10(x^4 + 2x^3 + 3x^2 + 2x + 1)$$

$$= 10\left(x^2 + \frac{1}{x^2} + 2\left(x + \frac{1}{x}\right) + 3\right)$$

$$= 10\left(\left(x + \frac{1}{x}\right)^2 + 2\left(x + \frac{1}{x}\right) + 1\right)$$

$$= 10\left(\left(x + \frac{1}{x}\right) + 1\right)^2 > 0; \forall x \in R$$

$\therefore f(x)$ is strictly increasing function. Since it is an odd degree polynomial it will have exactly one real root.

$$\text{Now, by observation } f(-1) = 3 > 0$$

$$f(-2) = -64 + 80 - 80 + 40 - 20 + 10$$

$$= -34 < 0$$

$$\Rightarrow f(x) \text{ has at least one root in } (-2, -1) \equiv (a, a+1)$$

$$\Rightarrow a = -2$$

$$\Rightarrow |a| = 2$$